

Question number	Answer	Additional guidance	Mark
5(a)	A description that includes four of the following points: • (action potential / impulse) causes {calcium ion channels to open / calcium ions to enter (presynaptic neurone)} (1) • (calcium ions cause) vesicles (containing neurotransmitter) to {fuse / bind} with presynaptic membrane (1) • which releases neurotransmitter (into the synaptic cleft) / neurotransmitter diffuses across synapse (1) • neurotransmitter binds with receptor (on post synaptic membrane) (1) • which initiates opening of sodium ion channels (resulting in an impulse in adjacent neurone) (1)	accept Ca^{2+}(voltage) gates open accept neurotransmitter released by exocytosis ignore cation channels accept Na^{+} (voltage) gates open	(4)

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5(b)	A description that includes four of the following points: • {isolate / identify} gene(s) for {enzymes / A and B} (from Madagascar periwinkle) (1) • cut plasmid with same {endonuclease / restriction enzyme} (1) • (genes are) {inserted into / attached to} plasmid (1) • sticky ends are joined using ligase (enzyme) (1) • (plasmid) which is taken up by the {<i>E. coli</i> / bacterium} (1) • <i>E. coli</i> {grown in (bio)fermenter / cultured} (1)	accept {isolate / cut (out)} {gene(s) / A and B} using restriction enzyme ignore {base sequence / DNA} unqualified ignore vector ignore vector accept join A and B and plasmid using ligase accept vector taken up by {<i>E. coli</i> / bacterium} accept bacteria given heat shock ignore gene gun	(4)

Question number	Answer	Additional guidance	Mark
5(c)	An answer that includes four of the following points: - serotonin concentration increases (over the 50 hours / time) (1) - tryptophan concentration decreased (over the 50 hours / time) (1) (because) tryptophan {converted to serotonin / is the (first) substrate} (1) - by the enzymes produced from the genes that have been inserted into the E coli (1) - conversion slows (after 30 hours) due to {fewer enzyme-substrate complexes / suitable stated limiting factor} (1)	accept there is no serotonin at the start accept by enzymes A and B suitable stated limiting factor e.g. substrate concentration / enzyme concentration / product inhibiting the enzyme etc ignore comments regarding error bars As tryptophan concentration decreases the serotonin concentration increases = 2 marks	(4)

